

Building Human-Centered AI Applications

Lab 05 – The “What to Wear” Bot

Objective: Create a tool that retrieves weather information based on a city name and provides clothing suggestions and advice for weather-appropriate outdoor activities

Part A: Define a weather data function

1. Create Lab5.py and incorporate it into your app’s navigation as a new Page
2. Visit wttr.in and try a few locations in your browser — wttr.in/Syracuse, NY wttr.in/Paris — to see what the service returns. No API key or account is required.
3. Create a function named `get_current_weather` that takes a location name and returns weather information. Request JSON from wttr.in by adding `?format=json` to the URL. The example below extracts two values just to confirm the call works — you’ll decide what your bot actually needs in Part B.

```
import requests
# location can be a city, a zip code, an airport code ('SYR'),
# or a landmark ('Eiffel+Tower')
# note: hard codes units to degrees Fahrenheit
def get_current_weather(location):
    url = f'https://wttr.in/{location}?format=json'
    response = requests.get(url, timeout=10)
    if response.status_code != 200:
        raise Exception(f'wttr.in error: status {response.status_code}')
    try:
        data = response.json()
    except ValueError:
        # unknown locations come back as plain text, not JSON
        raise Exception(f'Could not find a location named {location}')
    # json has three top-level sections:
    #   current_condition -- one entry, conditions right now
    #   weather           -- three entries, one per day, each with
    #                       min/max, astronomy, and hourly forecasts
    #   nearest_area      -- the location wttr.in actually matched
    current = data['current_condition'][0]
    # two examples; note that some values are nested one level deeper
    return {'location': location,
            'temperature': float(current['temp_F']),
            'description': current['weatherDesc'][0]['value']}
}
```

4. Test the function with ‘Syracuse, NY’ and ‘Lima, Peru’. Then open <https://wttr.in/Syracuse?format=json> directly in your browser and read through the full response so you know what’s available to you.

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Part B: Construct the “What to Wear” Bot

5. In Lab5.py, create a bot using Streamlit and OpenAI
 - a. Leverage code from a previous app that you think most appropriate
 - i. Note: This is **not a chatbot** — the user should input a city, and the bot should output the advice/suggestion
 - b. Define an LLM function/tool to get the weather, given a location
 - i. Decide which values your bot actually needs to give good clothing and activity advice and extend `get_current_weather` to return those. There are a lot of values you can use. Consider how the weather may change through the day and what users may need to know.
6. Provide this function/tool to the OpenAI API (i.e., as a tool)
 - a. Set the `tool_choice` to ‘auto’ (so it only asks for the weather when needed)
 - b. If you get a request for weather with no location, use ‘Syracuse, NY’ as the default
 - c. The tool will be invoked as needed
7. When this tool is involved, make another call to the OpenAI API:
 - a. Provide the weather information as part of the prompt to the LLM
 - b. Ask the LLM to provide suggestions on appropriate clothes to wear today and give suggestions for outdoor activities that are appropriate to the weather

Upload Lab5.py to Blackboard and provide the link to your application